

Man-Chen Liao

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EDUCATION

National Yang Ming Chiao Tung University (NYCU) B.S. in Electronics and Electrical Engineering <i>CGPA: 4.25/4.3; Last 60 GPA: 4.26/4.3; Class Ranking: 1/48</i>	Hsinchu, Taiwan Sept. 2024 – Present
National Yang Ming Chiao Tung University (NYCU) B.S. in Electrophysics (transferred to Electronics and Electrical Engineering in 2024 Fall)	Hsinchu, Taiwan Sept. 2023 – June 2024

RESEARCH EXPERIENCE

National Science and Technology Council (NSTC) <i>Undergraduate Researcher advised by Prof. Yu-Te Liao</i> - Awarded the NSTC College Student Research Scholarship for the project titled “An Integrated Electrochemical and Optical Smart Sensing Platform for Farm Animal Detection,” grant: 115-2813-C-A49-182-E (NTD 6,000/month). - Aimed to develop a dual-modal sensing platform for point-of-care animal disease detection. Focused on mitigating environmental interference and improving diagnostic reliability through data cross-validation.	Hsinchu, Taiwan July 2026 – Feb. 2027
Wireless Integrated Microsystems Lab, NYCU <i>Undergraduate Researcher advised by Prof. Yu-Te Liao</i> - Research on the design and optimization of a two-stage operational amplifier featuring a zero-nulling resistor (R_z) for frequency compensation. Aimed to achieve high phase margin, wide unity-gain bandwidth, and low power consumption. - Research on a new ultra-low noise Transimpedance Amplifier (TIA) topology with a DC current removal block to prevent integrator saturation. Aimed to achieve high transimpedance gain, wide bandwidth, and low input-referred noise under large background DC currents.	Hsinchu, Taiwan July 2025 – Present
Chien Lab, UC Berkeley <i>Visiting Student Researcher advised by Prof. Jun-Chau Chien</i> Underwent intensive training in advanced analog integrated circuits and data converters (ADC/DAC), and actively engaged in weekly journal club meetings to discuss cutting-edge IC research.	Berkeley, California July 2026 – Aug. 2026

SELECTED PROJECTS

CMOS Low-Dropout Regulator with Integrated Flash ADC <i>Semester Project, Analog Integrated Circuits Lab</i> - Constructed a transistor-level power management and data conversion circuit comprising an LDO, a Bandgap Reference (BGR), and a Flash ADC. Built the error amplifier (two-stage OP) for output voltage regulation and the BGR for stable 1.2V reference voltage, supporting digital output logic and a 100 mA load current. - Conducted rigorous pre- and post-layout simulations to verify loop stability, transient performance, and line/load regulation. Completed DRC, LVS and PEX using TSMC 180nm PDK.	Hsinchu, Taiwan Mar. 2026 – June 2026
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PUBLICATION

Man-Chen Liao and Hung-Chang Liao, “Design and Experimental Study of a Vocal Effects Unit Based on Frequency Response and Amplifier Design: A Comparison Between Practical Implementation and LTspice Simulation,” *24th Annual Hawaii International Conference on Education (HICE)*, Honolulu, Hawaii, USA, Jan. 2026.

TEACHING EXPERIENCE

Teaching Assistant, Electronics Laboratory (II) <i>Lecturer: Li-Chou Chen</i> Supported the instruction of the undergraduate laboratory course for over 50 students, helping refine course materials and prepare supporting instructional content.	Hsinchu, Taiwan Feb. 2026 – June 2026
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- Provided hands-on experimental guidance, debugged circuit setups, and delivered concept presentations during lab sessions.

HONORS & AWARDS

Shun-I Chu Zyxel Scholarship (NTD 120,000) Hsinchu, Taiwan
National Yang Ming Chiao Tung University 2026

Awarded to only Top 4 junior students selected across the College of Electrical and Computer Engineering.

Dean's List Award (4 Semesters) Hsinchu, Taiwan
National Yang Ming Chiao Tung University 2023 – 2025

Recognized for academic excellence, awarded to students ranked in the Top 5% of the class each semester.

The Datatronics Technology, Inc. Scholarship Hsinchu, Taiwan
National Yang Ming Chiao Tung University 2025

Awarded to only Top 1 sole recipient among all junior students in the Department of Electronics and Electrical Engineering.

NYCU Jin Qiu Bei Table Tennis Tournament Hsinchu, Taiwan
National Yang Ming Chiao Tung University 2025

Achieved 2nd Place in Women's Singles out of all participating competitors in the campus-wide tournament.

EXTRACURRICULAR ACTIVITIES

Hot Music Club, High School Taichung, Taiwan
President & Keyboardist Sept. 2021 – June 2022

- Led club operations, coordinated rehearsals and performances, and managed event planning.

NYCU EE Department Table Tennis Team Hsinchu, Taiwan
Member Sept. 2024 – Present

- Contributed to the team in winning the Annual Xiaomeichu Tournament, jointly organized by the EE departments of NYCU and NTHU, for two consecutive years.

NYCU Guitar Club Hsinchu, Taiwan
Member Sept. 2023 – Present

- Performed on stage as a vocalist, guitarist, and keyboardist in club performances.

2026 Virtual U.S.-Taiwan-Japan Semiconductor Collaboration Program Online
Participant 2026

- Completed the program hosted by the American Institute in Taiwan (AIT), focusing on semiconductor policies, technological innovation, and talent development.

TOOLS

Virtuoso, Spectre, Calibre, Hspice, Verilog-A, LTspice, Oscilloscope, Function Generator, Python, C++